A REMARK ON THE RANK THRESHOLD OF THE LATENT ALPHABET GADGET IN *THE UNIQUE GAMES THEOREM*

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Abstract. Lemma 3.1 of OpenAI's preprint *The Unique Games Theorem* constructs, for every $p > 0$, a noise gadget with nonlinear change probability at most p that is detected with probability at least $1/8$ by every linear family whose restriction to the output space has rank at least a threshold r_*; the preprint only asserts that r_* exists. When its proof is made explicit, the threshold it certifies is about p^{-16} (about $p^{-23.1}$ with its effective recipe). We show that the *same* construction satisfies the lemma with $r_* \le \lceil 1.5\, p^{-3} \rceil + 1$, and that every gadget satisfying the lemma has $r_* \ge 1/(4p)$. The two changes are bookkeeping: the final enlargement step, which squares the threshold, is unnecessary, and the paper's harmonic potential with a Markov bound is replaced by an exponential supermartingale.

Attribution and scope. The gadget, its construction and its analysis are OpenAI's work: family 102 of `https://github.com/openai/math` (commit `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`), preprint [1], Section 3 (`build/sections/02-latent.tex`). We change only the final packaging and the choice of parameters in the rank-survival argument, and add an elementary lower bound. We take the correctness of the preprint's lemmas as given. The community presentation [2] already takes the output space equal to the block output (for ℓ divisible by 3) and so avoids the enlargement step; we acknowledge it below. Developed with AI assistance (Anthropic Claude, Opus 5.5) under [AUTHOR NAME]'s direction; the independent adversarial review was run by a separate AI session and is not human peer review.

1. The quantity

Following [1, Lemma 3.1], for $p \in (0,1)$ let $r_*(p)$ be the least integer r such that for every sufficiently large ℓ there are a finite binary space $\mathcal{V} \supseteq K = \mathbb{F}_2^\ell$, a map $C : \mathcal{V} \to K$ with $C(x+k) = C(x) + k$, and a law μ on $\mathcal{V}$ with $\Pr_{x,a\sim\mu}[C(x+a) \ne C(x)] \le p$, such that every $G \le \mathcal{V}^*$ with $\dim(G|_K) \ge r$ satisfies $\Pr_{a\sim\mu}[\exists g \in G :\ g(a) \ne 0] \ge \frac{1}{8}$.

2. Lower bound

Proposition 2.1. *$r_*(p) \ge 1/(4p)$. More precisely, any such gadget with change probability p and any $r \le \ell$ for which the detection property holds satisfy $r\,p'/2 \ge \frac{1}{8}\prod_{i<r}\bigl(1 - \frac{2^i-1}{2^\ell-1}\bigr)$, where $p' = p\,2^\ell/(2^\ell-1)$.*

Proof. For $\sigma \in K^*$ put $F_\sigma(x) = (-1)^{\sigma(C(x))}$. Equivariance gives $F_\sigma(x+k) = (-1)^{\sigma(k)}F_\sigma(x)$, so its Fourier coefficient $\hat{F}_\sigma(h)$ vanishes unless $h|_K = \sigma$; by Parseval $\nu_\sigma(h) = \hat{F}_\sigma(h)^2$ is a probability law on $\mathcal{V}^*$. With $\hat{\mu}(h) = \mathbb{E}_a(-1)^{h(a)}$, $s_\sigma := \mathbb{E}_{x,a}F_\sigma(x)F_\sigma(x+a) = \sum_h \nu_\sigma(h)\hat{\mu}(h)$, and $\Pr[C(x+a) = C(x)] = \mathbb{E}_{\sigma\in K^*}s_\sigma$. Since $s_0 = 1$, $\mathbb{E}_{\sigma\ne 0}(1-s_\sigma) \le p'$, and $1 - s_\sigma = 2\mathbb{E}_{h\sim\nu_\sigma}\Pr_a[h(a) = 1]$. Draw $\sigma_1, \dots, \sigma_r$ independently uniform in $K^* \setminus 0$, then $h_i \sim \nu_{\sigma_i}$, and let $G = \operatorname{span}(h_i)$. Detection by G means $h_i(a) = 1$ for some i, so by the union bound $\mathbb{E}[\Pr_a(\text{detect})] \le rp'/2$. On the event that the σ_i are linearly independent, $\dim(G|_K) = r$ because $h_i|_K = \sigma_i$, and detection is at least $\frac{1}{8}$. Letting $\ell \to \infty$ with $r = r_*$ fixed gives $r_* \ge 1/(4p)$. □

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3. Upper bound for the same construction

We use the notation of [1, §3]: $F = \mathbb{F}_{2^d}$, $B = F^3$ of binary dimension $b = 3d$, $\theta = 1/(q^2+q+1)$ with $q = 2^d$, the height-s tree P_s, its noise ξ_s, the quotient $(V_0, B, \overline{C}, \nu)$, lifts, the filtration $\mathcal{F}_j$, and the indicator B_j that the current domain S_j is nongeneric of positive rank.

Theorem 3.1. *$r_*(p) \le \lceil 1.5\,p^{-3}\rceil + 1$ for every $p \in (0,1)$; for output dimensions ℓ divisible by 3 the threshold $\lceil 1.5\,p^{-3}\rceil - 1$ suffices.*

Parameters. $r_0 = \lceil 1.5\,p^{-3}\rceil - 1 \ge 1$, $\varepsilon_0 = 1/\big(120(\ln(1/p)+1)2^{r_0}\big)$, $d \ge d_0(p)$ such that $3d \ge r_0$, $\theta \le 0.01$, $2^{r_0}\theta \le 1/120$, and a uniform rank-r subspace of B^* is nongeneric with probability at most ε_0 for each $1 \le r \le r_0$, the last condition being verified through the explicit union bound in the proof of [1, Lemma 3.6], $\big[\binom{2^r-1}{2}q^{-2} + \binom{2^r-1}{3r+1}(6r+3)q^{-1}\big]/\big(1-(2^r-1)q^{-3}\big) \le \varepsilon_0$. That bound and the other three conditions are monotone in d, so they persist for all $d \ge d_0$. (The true nongeneric fraction need not be monotone in d; this is why we use the bound.) Finally $s = \lceil \ln(1/p)/\theta \rceil$. None of r_0, ε_0 depends on d.

Proof. Change probability. By [1, Lemma 3.7], $p_s \le (1-\theta)^s \le e^{-\theta s} \le p$.

Detection for $\ell = 3d$, $K = B$. Let $G_0 \le V_0^*$ with $\dim(G_0|_B) \ge r_0$. Choose an r_0-dimensional $S \le G_0|_B$ and a linear section $\sigma : S \to G_0$ of restriction; then $\psi_z = \sigma(z) \circ q_0$ is a lift on P_s with domain S in the sense of [1, eq. (3.16)], exactly as in the paper's argument after [1, Prop. 3.9], except that the domain is S rather than all of B^*. Follow the random leaf path as in the proof of [1, Prop. 3.9], starting from $S_0 = S$ (not necessarily generic) and producing S_j of rank r_j. The paper's estimates give, conditionally on $\mathcal{F}_j$: if S_j is generic then $r_j - r_{j+1} \le 1$ and $\Pr[r_{j+1} < r_j] \le 3r_j\theta$; for every S_j, $\Pr[r_{j+1} < r_j] \le 2^{r_j}\theta$; and $\mathbb{E}[B_{j+1} \mid \mathcal{F}_j, A_j] \le \varepsilon_0$.

Call step j *bad* if a loss occurs at step j and either $j = 0$ or $B_j = 1$. Then

$$\Pr[\text{some step } j < s \text{ is bad}] \le 2^{r_0}\theta + s\theta\, 2^{r_0}\varepsilon_0 \le \tfrac{1}{120} + \tfrac{\ln(1/p)+\theta}{120(\ln(1/p)+1)} \le \tfrac{1}{60}.$$

Let $\tilde r_j$ equal r_j up to the first bad step and stay constant afterwards, and $W_j = e^{-3\theta j}/(\tilde r_j + 1)$. At a step $j \ge 1$ before the first bad step with S_j generic and $r = r_j \ge 1$, a loss of exactly one occurs with conditional probability $\pi \le 3r\theta$, so

$$\mathbb{E}\Big[\tfrac{1}{\tilde r_{j+1}+1} \;\Big|\; \mathcal{F}_j\Big] = \tfrac{1}{r+1} + \tfrac{\pi}{r(r+1)} \le \tfrac{1+3\theta}{r+1} \le \tfrac{e^{3\theta}}{r+1};$$

at a nongeneric step, or at $j = 0$, $\tilde r_{j+1} = \tilde r_j$ by construction, and $r = 0$ is absorbing. Hence (W_j) is a supermartingale and

$$\Pr[r_s = 0,\ \text{no bad step}] \le e^{3\theta s}\mathbb{E}W_s \le \frac{e^{3\theta s}}{r_0+1} \le \frac{p^{-3}e^{3\theta}}{1.5\,p^{-3}} \le \tfrac{2}{3}e^{0.03} < 0.687,$$

using $\theta s \le \ln(1/p) + \theta$. So $\Pr[r_s = 0] < 0.687 + 1/60 < 0.704 < 3/4$. On survival, the leaf family of rank $r_s \ge 1$ detects the independent uniform leaf symbol with probability at least $\frac12$ [1, proof of Prop. 3.9], and such a detection is a detection of ξ_s by the original family. Hence G_0 detects ν with probability at least $\frac14 \cdot \frac12 = \frac18$.

Other ℓ. For $\ell = 3d + e$, $e \in \{1,2\}$, $d \ge d_0(p)$, put $\mathcal{V}' = V_0 \oplus \mathbb{F}_2^e$, $K' = B \oplus \mathbb{F}_2^e$, $C'(x,y) = (\overline{C}(x), y)$, and noise $\nu \otimes \delta_0$. Equivariance and the change probability are inherited. If $G \le \mathcal{V}'^*$ has $\dim(G|_{K'}) \ge r_0 + 2$, the projections g_V of its elements to V_0^* form a space with $\dim(\cdot|_B) \ge r_0$, and $g(a,0) = g_V(a)$; so G detects with probability at least $\frac18$. □

Remark 3.2 (where the original loses).

(1) *Enlargement* ([1], proof of Lemma 3.1, lines 674–745). The paper detects only families whose restriction is all of B^*, then enlarges B to an arbitrary K via injections, with $r_* = b + 2 = 3d + 2$. Since d must exceed about $3r_0^2$ (the union bound in [1, Lemma 3.6]), this

squares the threshold: $r_* \approx 9r_0^2$. The descent above works from any r_0-dimensional domain, so $r_* = r_0$ (+2 for padding). For $\ell \in 3\mathbb{Z}$ this simplification already appears in [2].

(2) *Harmonic potential and Markov* ([1], lines 534–612). The paper bounds the expected harmonic drop by 4θ per level and applies Markov's inequality at $\frac{1}{2}$, so it needs $s\theta \le H_{r_0}/8$ and hence $H_{r_0} \gtrsim 8\ln(1/p)$, i.e. $r_0 \approx e^{8-\gamma}p^{-8}$ (exponent $8 = 4 \times 2$). The supermartingale $e^{-3\theta j}/(r_j + 1)$ uses the per-level loss rate $3r\theta$ directly and gives $r_0 \approx 1.5\,p^{-3}$.

(3) *Limit of this route.* For the extremal chain allowed by the estimate $\Pr[\text{loss}] \le 3r\theta$ (loss exactly one with probability $3r\theta$), the extinction rank after s steps is close to $\mathrm{Bin}(r_0, p^3)$, and the detection probability is then about $\mathbb{E}[1 - 2^{-r_s}] \approx 1 - e^{-r_0 p^3/2}$; so an argument that uses only this estimate (and the leaf bound) needs $r_0 \ge (2\ln\frac{8}{7} + o(1))p^{-3} \approx 0.267\,p^{-3}$. Our constant 1.5 is within a factor 5.6 of this. This is a limit of the proof method only, not a lower bound on r_*: $3r\theta$ is an upper bound on the loss rate. Exact computations of the chain are in the scripts.

4. Numbers

The thresholds certified by each argument (the paper's union bound is used to fix d where needed):

	$p = 10^{-1}$	$p = 10^{-2}$	$p = 10^{-4}$	$p = 10^{-8}$
paper, main text ($r_* = 3d + 2$)	$2^{77.7}$	$2^{130.9}$	$2^{237.2}$	$2^{449.8}$
paper, effective recipe	$2^{116.9}$	$2^{186.2}$	$2^{347.8}$	$2^{647.8}$
harmonic analysis, no enlargement	$2^{37.3}$	$2^{63.9}$	$2^{117.0}$	$2^{223.3}$
Theorem 3.1	$2^{10.6}$	$2^{20.5}$	$2^{40.4}$	$2^{80.3}$
lower bound $1/(4p)$	$2^{1.3}$	$2^{4.6}$	$2^{11.3}$	$2^{24.6}$

Asymptotically the exponents of $1/p$ are 16 (main text), $16/\ln 2 \approx 23.1$ (effective recipe), 8, 3, and 1. The truth for this gadget lies between p^{-1} and p^{-3}.

What is not claimed

We do not claim the order of $r_*(p)$; the gap between p^{-1} and p^{-3} remains. We do not claim any change to the Unique Games reduction beyond the size of its alphabet parameters (which depend on r_* through $\eta = 2^{-r_*}/16$ in later sections); in particular the final alphabet remains astronomically large. The paper's effective algorithm picks d by exact enumeration and could certify smaller d than the union bound; our "before" numbers are the thresholds certified by the paper's own proof. No Lean formalisation.

References

[1] OpenAI, *The Unique Games Theorem*, September 23, 2026, https://github.com/openai/math (family 102), Section 3.

[2] `lgboim/unique-games-short-proof`, *The Unique Games Theorem: an 11-page self-contained proof and a faithful presentation of OpenAI's proof*, GitHub, October 9, 2026.